



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
Constituent of Symbiosis International (Deemed University), Pune

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Subject: Programming with Java	Sem: III AY: 2026-27
Assignment No: 5	
TITLE: Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.	

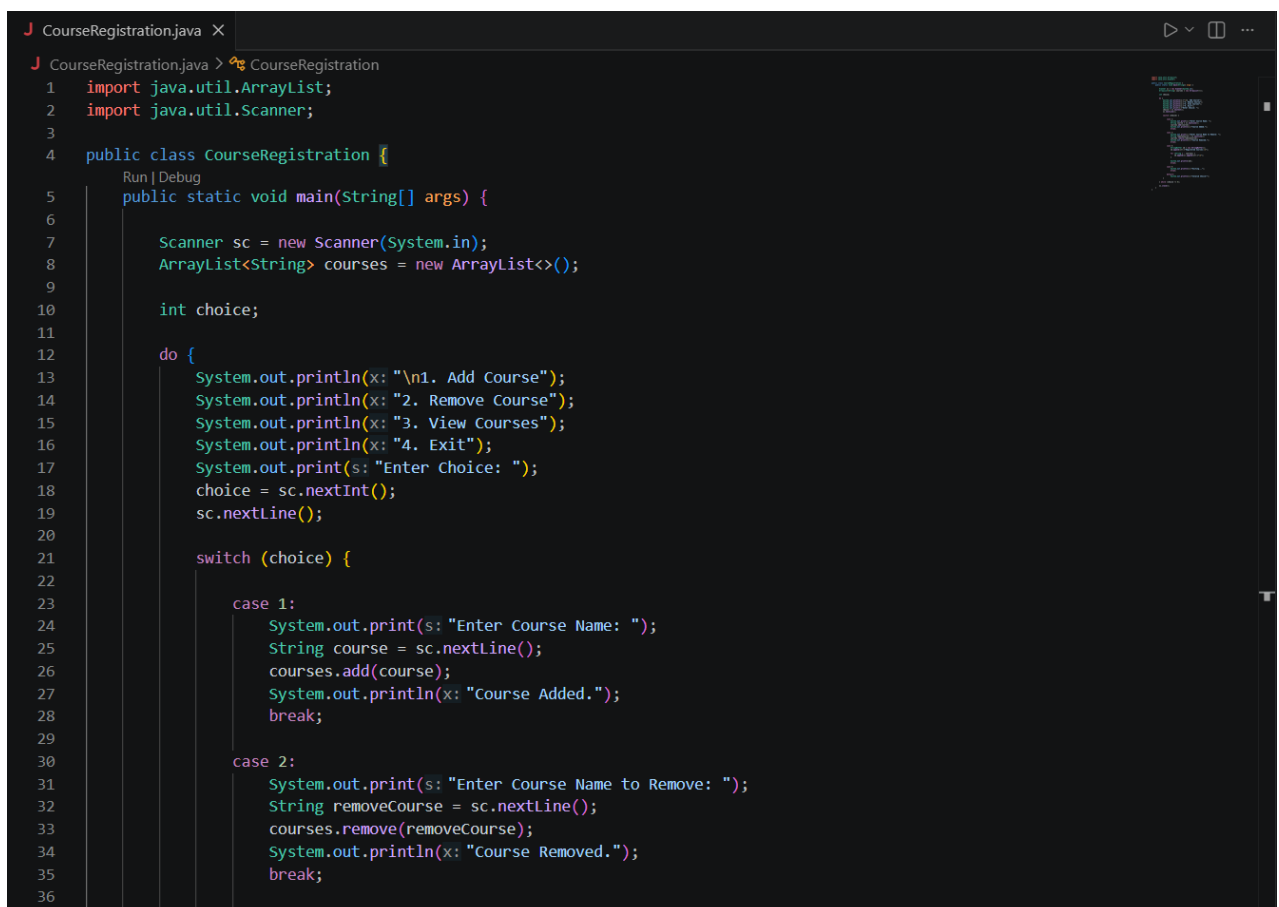
1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

```
J TodoList.java >  TodoList
1  import java.util.ArrayList;
2
3  public class TodoList {
4      Run | Debug
5      public static void main(String[] args) {
6
7          ArrayList<String> tasks = new ArrayList<>();
8
9          // Adding tasks
10         tasks.add(e: "Complete Java Assignment");
11         tasks.add(e: "Study for Exam");
12         tasks.add(e: "Submit Project");
13
14         // Display tasks using StringBuffer
15         StringBuffer sb = new StringBuffer();
16         sb.append(str: "To-Do List:\n");
17
18         for (int i = 0; i < tasks.size(); i++) {
19             sb.append((i + 1) + ". " + tasks.get(i) + "\n");
20         }
21
22         System.out.println(sb);
23     }
}
```

OUTPUT:

```
PS C:\Users\Arpit\java_assignments> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetails  
InExceptionMessages' '-cp' 'C:\Users\Arpit\AppData\Roaming\Code\User\workspaceStorage\4bc4a89ca2bf606057b81e856df867b6\redhat.j  
ava\jdt_ws\java_assignments_df3da72d\bin' 'ToDoList'  
To-Do List:  
1. Complete Java Assignment  
2. Study for Exam  
3. Submit Project
```

2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.



```
J CourseRegistration.java X
J CourseRegistration.java > CourseRegistration
1 import java.util.ArrayList;
2 import java.util.Scanner;
3
4 public class CourseRegistration {
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8         ArrayList<String> courses = new ArrayList<>();
9
10        int choice;
11
12        do {
13            System.out.println(x: "\n1. Add Course");
14            System.out.println(x: "2. Remove Course");
15            System.out.println(x: "3. View Courses");
16            System.out.println(x: "4. Exit");
17            System.out.print(s: "Enter Choice: ");
18            choice = sc.nextInt();
19            sc.nextLine();
20
21            switch (choice) {
22
23                case 1:
24                    System.out.print(s: "Enter Course Name: ");
25                    String course = sc.nextLine();
26                    courses.add(course);
27                    System.out.println(x: "Course Added.");
28                    break;
29
30                case 2:
31                    System.out.print(s: "Enter Course Name to Remove: ");
32                    String removeCourse = sc.nextLine();
33                    courses.remove(removeCourse);
34                    System.out.println(x: "Course Removed.");
35                    break;
36            }
37        } while (choice != 4);
38    }
39 }
```

```

37         case 3:
38             StringBuffer sb = new StringBuffer();
39             sb.append(str: "Registered Courses:\n");
40
41             for (String c : courses) {
42                 sb.append(c).append(str: "\n");
43             }
44
45             System.out.println(sb);
46             break;
47
48         case 4:
49             System.out.println(x: "Exiting...");
50             break;
51
52         default:
53             System.out.println(x: "Invalid Choice!");
54     }
55
56     } while (choice != 4);
57
58     sc.close();
59 }
60 }

```

OUTPUT:

```

PS C:\Users\Arpit\java_assignments> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetails
InExceptionMessages' '-cp' 'C:\Users\Arpit\AppData\Roaming\Code\User\workspaceStorage\4bc4a89ca2bf606057b81e856df867b6\redhat.j
ava\jdt_ws\java_assignments_df3da72d\bin' 'CourseRegistration'

1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice: 1
Enter Course Name: Java
Course Added.

1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice: 1
Enter Course Name: PYTHON
Course Added.

1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice: 3
Registered Courses:
Java
PYTHON

1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice: 2
Enter Course Name to Remove: Java
Course Removed.

```